

## Scalability & Future Plan

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|----------------------|---------------|
| <b>Date</b>          | 1 July 2026   |
| <b>Team ID</b>       |               |
| <b>Project Name</b>  | Rising Waters |
| <b>Maximum Marks</b> | 1 Mark        |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                                                                            | Impact                                              | Priority to Address<br>(High / Medium / Low) |
|------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------|
| 1    | Uses only a historical dataset ( <code>flood_data.csv</code> ) without real-time weather information. | Prediction accuracy depends on historical data.     | High                                         |
| 2    | No dashboard or map visualization.                                                                    | Users cannot visualize flood-prone areas or trends. | Medium                                       |
| 3    | Application runs only on localhost.                                                                   | Cannot be accessed remotely by multiple users.      | High                                         |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                                                                             | Proposed Upgrade / Solution                                                           |
|------|--------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1    | User Load          | Deployed on <b>Render</b> and supports online access                                                      | Enable auto-scaling and load balancing to handle a larger number of concurrent users. |
| 2    | Data Storage       | Uses a local CSV dataset and saved model files ( <code>best_model.pkl</code> , <code>scaler.pkl</code> ). | Integrate a cloud database such as MySQL or PostgreSQL for scalable data management.  |

|   |             |                                                                      |                                                                                 |
|---|-------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 3 | Performance | Predictions are generated using the deployed machine learning model. | Optimize model performance and use cloud infrastructure for faster predictions. |
| 4 | Security    | No authentication or user management.                                | Implement user authentication, role-based access, and secure data validation.   |

Future Roadmap:

| Phase   | Planned Feature / Enhancement                                                                                   | Target Timeline | Expected Impact                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------|
| Phase 2 | Integrate real-time weather APIs, rainfall data, and river water-level monitoring.                              | Next 3 Months   | Improve prediction accuracy and provide real-time flood alerts.   |
| Phase 3 | Implement user authentication, location-based notifications, SMS/Email alerts, and a mobile-friendly interface. | Next 6 Months   | Enable users to visualize flood-prone regions and monitor trends. |